

ORDER FULFILLMENT PERFORMANCE ANALYTICS

Executive Summary

Portfolio Project 1 · Supply Chain Analytics · 2026

Business Problem

A supply chain operation lacks centralized visibility into fulfillment performance. Without a unified analytical view, service failures, profitability leaks, and structural inefficiencies remain undetected. This project builds an end-to-end analytics solution — from raw transactional data to a decision-oriented dashboard and statistical analysis — to surface actionable operational insights.

Approach

180,519 order records (2015–2018) were transformed from a raw denormalized dataset into a validated star schema dimensional warehouse (PostgreSQL), connected to a Power BI Control Tower dashboard, and explored through a Python EDA notebook (pandas, seaborn, statsmodels). The pipeline covers data audit, dimensional modeling, warehouse validation, business intelligence, and statistical analysis.

Key Findings

- **54.83% late delivery rate / 45.17% OTIF:** More than half of all orders carry a late delivery risk flag. First Class — the premium tier — is the worst performer at 95.32%, driven by a rigid 2-day promise with zero operational flexibility. Standard Class is the most reliable at 38.07%.
- **18.7% of orders are loss-making:** Nearly 1 in 5 orders generates negative profit, a finding invisible in the aggregate 10.78% margin. Discount rate does not explain the losses — some orders lose money with zero discount applied, pointing to cost structures not captured in the available data.
- **Four shipping tiers are operationally two:** Delivery time analysis reveals a bimodal distribution: First Class delivers in exactly 2 days (no variation), while Standard Class and Second Class are statistically indistinguishable (2–6 days, median 4 days). The four-tier product structure does not reflect operational reality.
- **Discount erodes margin in aggregate but not at row level:** Power BI confirms that average profit margin decreases as discount rate increases. However, correlation analysis and OLS regression both show near-zero relationship at the individual order level ($r = -0.003$, $R^2 = 0.000$), indicating that multiple interacting factors dominate at row level.

Recommendations

- **Simplify the shipping tier structure:** Consolidate four shipping modes into two tiers (express and standard) that reflect actual delivery capabilities. Reset customer expectations on First Class to reduce the structural late delivery rate.
- **Investigate loss-making orders:** 18.7% unprofitable orders require a cost-level investigation beyond what the current dataset supports. Capturing shipping cost, return rates, and handling costs per order is a prerequisite for root cause analysis.
- **Invest in data completeness:** Key planned KPIs (Shipping Cost Ratio, Perfect Order Rate) could not be calculated due to missing data. Collecting shipping cost, order completeness, and damage data would unlock a significantly richer analytical model.

Data Limitations

The dataset covers 2015–2017 fully and Q1 2018 only (incomplete year). No shipping cost or order completeness data is available. Risk analysis is limited to the late_delivery_risk binary flag. All findings should be interpreted within these constraints.